REPORT

OF

DIABETES DIET ADVISOR

A Web-Based Health and Wellness Application

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE (AI & ML)

By

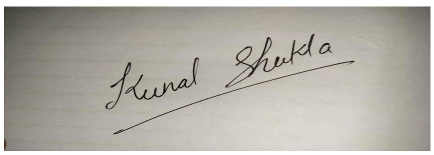
KUNAL SHUKLA

Field	Details
Enrollment Number	25SCS1003003291
Program	B.Tech CSE (AI & ML)
Batch	2025-2029
Semester / Section	3rd Semester / 2CSE10
Internship	AI-Driven Web Application & Product Development
Organization / Program	Lenovo LEAP NextGen Scholar Program; implemented by BharatCares in association with AICTE
Internship Period	15 June 2026 - 30 July 2026

CANDIDATE'S DECLARATION

I, KUNAL SHUKLA, do hereby declare that the internship report titled "Diabetes Diet Advisor" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI & ML). All references used in preparing this report have been acknowledged appropriately. I affirm that this report presents my internship learning and project work and has not been submitted to any other institute for any degree or diploma requirement.

I understand that I am responsible for the authenticity of the work presented and for avoiding copyright infringement or misrepresentation of material.



Signature: ____________________________

Student Name: KUNAL SHUKLA

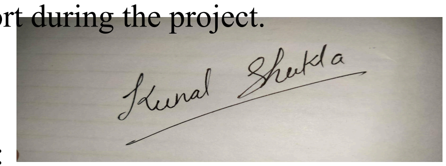
Enrollment No.: 25SCS1003003291

ACKNOWLEDGEMENT

I express my sincere gratitude to everyone who supported me throughout the six-week AICTE internship on AI-Driven Web Application & Product Development under the Lenovo LEAP NextGen Scholar Program. The internship was an initiative of Lenovo, implemented by BharatCares in association with the All India Council for Technical Education (AICTE).

I am thankful to the trainers, mentors and program coordinators for the structured masterclasses, mentoring sessions, learning resources, assessment guidance and career-readiness activities. The internship provided practical exposure to web application development, product thinking, prompt engineering and project-based learning.

I also thank the School of Computer Science and Engineering, IILM University, Greater Noida, for providing the academic framework for summer internship learning and submission. Finally, I express gratitude to my parents and peers for their encouragement and support during the project.



Signature: ______________________

KUNAL SHUKLA

INTERNSHIP COMPLETION CERTIFICATE

The internship completion certificate issued under the Lenovo LEAP NextGen Scholar Program records that Kunal Shukla successfully completed the six-week AICTE internship on "AI-Driven Web Application & Product Development" during June-July 2026. The program was an initiative of Lenovo, implemented by BharatCares, and conducted in association with AICTE.

Certificate details supplied by the student:
Unique ID: 6955775
Certificate Date: 24 July 2026

Field	Details
Certificate Holder	Kunal Shukla
Unique ID	6955775
Date	24/07/2026
Program	Lenovo LEAP NextGen Scholar Program
Internship	AI-Driven Web Application & Product Development

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4. PROJECT DESCRIPTION

Project Title: Diabetes Diet Advisor (DiaDiet)

Diabetes Diet Advisor, branded in the interface as "DiaDiet", is a browser-based student project designed to provide simple food-choice guidance and wellness utilities for people interested in diabetes-aware lifestyle management. The project is aligned with United Nations Sustainable Development Goal 3 - Good Health and Well-being.

The application presents a clean single-page interface with navigation to the food advisor, healthy diet tips, BMI calculator, water-intake tracker and project information. The core advisor accepts a blood sugar value and a selected food or fruit and then displays a simple recommendation. Additional modules encourage awareness of hydration, healthy weight and everyday lifestyle practices.

The project was developed as part of the learning outcomes of the six-week AICTE internship on AI-Driven Web Application & Product Development. The internship itself emphasized practical product development, AI-oriented thinking, structured prompting and hands-on web application learning. The current DiaDiet implementation is accurately described as a front-end web application with rule-based interactive logic; no trained machine-learning model or external generative-AI API is claimed in this report.

Field	Details
Live Website	https://the-coder6.github.io/diabetes-diet-advisor/#advisor
Source Repository	https://github.com/the-coder6/diabetes-diet-advisor.git
SDG Alignment	UN SDG 3 - Good Health and Well-being
Application Type	Responsive front-end web application

4.1 INTRODUCTION

Digital health tools can make educational information easier to access and understand. Diet is an important part of diabetes awareness, yet users may find it difficult to translate general dietary advice into simple day-to-day choices. The Diabetes Diet Advisor project explores how a lightweight web interface can organize basic food guidance and wellness tools in one place.

The project does not attempt to diagnose diabetes, prescribe treatment or replace a qualified medical professional. Its purpose is educational: to demonstrate web development, user-interface design, JavaScript-based interactivity and responsible presentation of health-oriented information.

The internship orientation described AI-driven product development as a practical, product-focused learning journey rather than only a lecture series. It also introduced the Sustainable Development Goals as a framework for building socially relevant projects. DiaDiet follows that product-development spirit by selecting a health and well-being problem and converting it into a working, publicly deployed web application.

The application is designed around simplicity. A user can move directly from the home section to the advisor, enter a blood sugar level, select a food item and receive an immediate message. The same page also contains supporting features such as healthy diet tips, BMI calculation and a water-intake reminder.

4.1.1 Internship Context

The Lenovo LEAP internship ran as a six-week AICTE internship in AI-Driven Web Application & Product Development. Program resources document an orientation, Masterclasses 1-7, mentoring sessions, completion guidance, Skill India assessment scheduling, career-readiness sessions and a hackathon orientation. The student reports completing the Skill India assessment, although a separate assessment certificate was not available for attachment at the time this report was prepared.

4.2 ORGANIZATION PROFILE

The internship was offered under the Lenovo LEAP NextGen Scholar Program. The student's offer letter identifies the program as a Lenovo initiative, implemented by BharatCares in association with AICTE. The internship was virtual and focused on AI-Driven Web Application & Product Development.

The official internship resources describe a structured learning journey supported by masterclasses, mentoring and project-guided sessions. Program materials also provided recordings, trainer decks and orientation decks for review. The orientation emphasized practical product development and described AI-driven web applications as products that can integrate AI/ML capabilities to enhance experience, automate tasks or generate intelligent outputs.

BharatCares served as the implementation partner for the program. AICTE was the associated technical-education body. The completion certificate supplied by the student similarly identifies Lenovo, AICTE and BharatCares and confirms successful completion of the six-week internship.

4.2.1 Internship Schedule

Field	Details
15 Jun 2026	Orientation + Master Class 0
17 Jun	Masterclass 1
22 Jun	Masterclass 2
24 Jun	Masterclass 3
26 Jun	Masterclass 4
29 Jun	Mentoring Session
1 Jul	Masterclass 5
3 Jul	Masterclass 6
6 Jul	Masterclass 7
8 Jul	Completion Guidance / Mentoring
10 Jul	Skill India Assessment
11 & 13 Jul	Career Readiness Sessions
15 Jul	Hackathon Orientation

4.3 PROBLEM STATEMENT

People looking for simple diabetes-aware dietary guidance often encounter information that is scattered across multiple websites, written in technical language, or disconnected from everyday decisions. A student-level web solution can demonstrate how basic dietary awareness and wellness utilities may be brought together in an accessible interface.

The project therefore addresses the following development problem: how can a lightweight, responsive web application provide quick food-choice guidance while also encouraging basic wellness habits such as hydration, weight awareness and healthier daily routines?

The design problem includes several sub-problems: keeping the interface easy for non-technical users; accepting simple user inputs; generating immediate and understandable feedback; presenting additional lifestyle tips without clutter; and making the application accessible through a public web link without requiring installation.

Because the project concerns health, another important requirement is responsible scope. The application is an educational demonstration and not a clinical decision-support system. Its recommendations should not be interpreted as individualized medical advice.

4.3.1 User Need

- Simple and fast food guidance interface.
- Readable feedback after user input.
- Basic wellness utilities in the same application.
- Responsive access through a browser.
- Clear educational rather than diagnostic purpose.

4.4 PROJECT OBJECTIVES

The primary objective of the Diabetes Diet Advisor is to apply internship learning to a complete, working web product aligned with a socially relevant Sustainable Development Goal.

- Design and develop a clean, responsive single-page web application.
- Create a food-advisor module that accepts blood sugar level and selected food/fruit as input.
- Provide immediate rule-based recommendation messages in understandable language.
- Include practical wellness modules such as healthy diet tips, BMI calculation and water-intake tracking.
- Use HTML for structure, CSS for presentation and JavaScript for client-side interaction.
- Deploy the completed application publicly using GitHub Pages.
- Maintain project source files in a public GitHub repository.
- Demonstrate product-development learning from the Lenovo LEAP internship.
- Align the project theme with UN SDG 3 - Good Health and Well-being.
- Document limitations accurately and avoid claiming unsupported AI/ML functionality.

4.4.1 Learning Objectives

The internship also supported broader learning objectives: understanding prompt engineering, improving structured problem-solving, learning the workflow from idea to working interface, receiving mentoring, completing assessment activities and gaining career-readiness exposure.

4.5 SCOPE OF THE PROJECT

The current scope of DiaDiet is intentionally focused on a browser-based educational prototype. It demonstrates how health-oriented information and interactive utilities can be organized into a coherent product without requiring a backend server or user account system.

Included in Current Scope

- Home/hero section with project purpose and SDG 3 label.
- Food recommendation form using blood sugar level and food/fruit selection.
- Immediate recommendation display.
- Healthy diet tips cards.
- BMI calculator using height and weight inputs.
- Water intake tracker with add/reset controls.
- About and contact section.
- Light/dark appearance control visible in the interface.
- Public deployment through GitHub Pages.

Outside Current Scope

- Medical diagnosis or treatment recommendations.
- Individualized diet plans based on complete clinical history.
- Real-time integration with glucose-monitoring devices.
- User authentication or cloud database.
- Doctor/patient dashboards.
- Trained machine-learning model inference.
- External generative-AI API integration.

These exclusions are important because they define the prototype honestly and provide a roadmap for future enhancement rather than overstating the current implementation.

4.6 TECHNOLOGIES AND TOOLS USED

Field	Details
HTML5	Defines page structure, navigation, forms, cards and content sections.
CSS3	Controls layout, typography, responsive styling, cards, buttons and visual states.
JavaScript	Implements interactive logic for recommendations, BMI calculation, water tracking and interface behavior.
GitHub	Hosts the public source-code repository and version history.
GitHub Pages	Publishes the static website as a live web application.
Web Browser	Used for testing and running the deployed application.
Prompt Engineering Learning	Covered in Lenovo LEAP Masterclass 1 as part of internship learning; not represented as a runtime AI service in DiaDiet.

4.6.1 Why This Stack Was Suitable

The selected stack is appropriate for a student prototype because it is lightweight, widely supported and deployable without server infrastructure. HTML, CSS and JavaScript are sufficient for the implemented functionality, while GitHub and GitHub Pages provide source management and public deployment. This also keeps the project easy to inspect during academic evaluation.

4.7 SYSTEM ARCHITECTURE

DiaDiet follows a client-side architecture. The user accesses the application in a web browser. HTML provides the content and input controls, CSS provides visual presentation, and JavaScript reads user actions, performs simple calculations or rule checks, and updates the page with results. The application is hosted as static files on GitHub Pages.

Logical Architecture

USER

↓

WEB BROWSER

↓

HTML USER INTERFACE

↓

JAVASCRIPT INTERACTION / RULE LOGIC

↓

RESULT DISPLAY

CSS PRESENTATION supports the interface

GitHub Pages hosts the static application

4.7.1 Module Architecture

Field	Details
Food Advisor	Input: blood sugar value + selected food; Output: recommendation message.
Diet Tips	Static educational wellness tips displayed as cards.
BMI Calculator	Input: weight and height; Output: calculated BMI and category.
Water Tracker	Tracks glasses toward a displayed daily goal during the page session.
About / Contact	Explains the project, creator and SDG alignment.

4.8 METHODOLOGY

The project followed a simple iterative product-development methodology suitable for the internship context.

Step 1 - Problem Selection

A health and wellness theme was selected and connected to UN SDG 3. The idea was narrowed to a diabetes diet guidance application that could be completed as a functional web prototype.

Step 2 - Requirement Identification

The essential user journey was defined: open the application, enter a blood sugar value, select a food item, request a recommendation and read the result. Supporting features were then added to improve the usefulness of the product.

Step 3 - Interface Design

The interface was organized into Home, Advisor, Tips, BMI, Water and About sections. A simple green health-oriented visual language was used, with cards and clear call-to-action buttons.

Step 4 - Front-End Implementation

HTML was used for structure, CSS for styling and JavaScript for interactive behavior. Each feature was tested in the browser during development.

Step 5 - Deployment and Verification

The project source was maintained on GitHub and deployed through GitHub Pages. Screenshots were captured from the live website to document the working modules.

4.8.1 IMPLEMENTATION - HOME PAGE

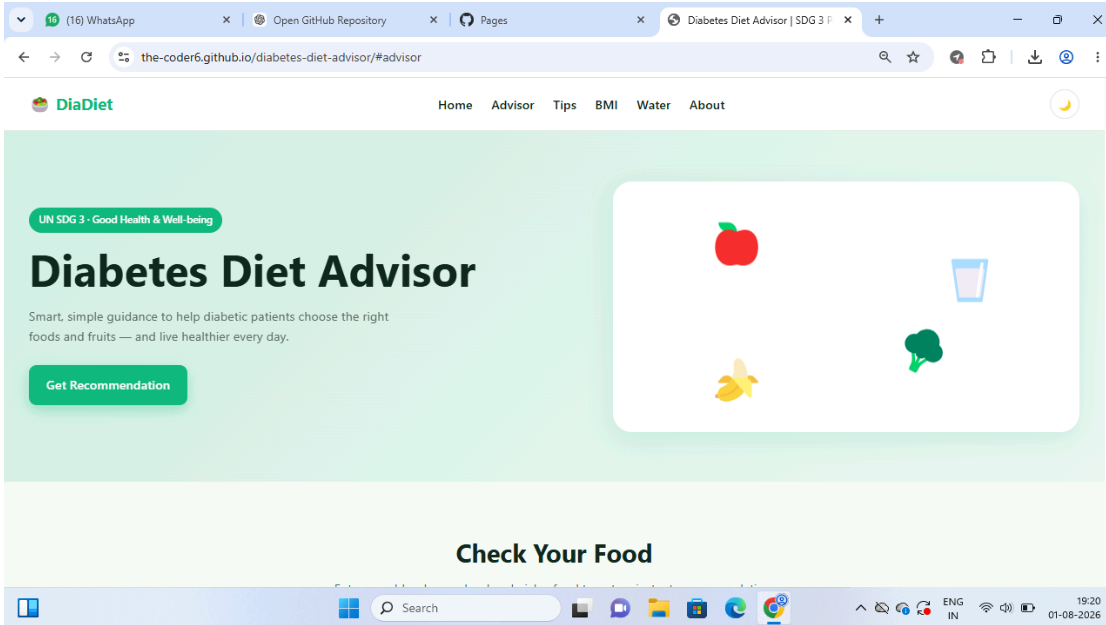

Figure 4.1: DiaDiet home page and project introduction

The home section introduces the Diabetes Diet Advisor and communicates the project's purpose in a concise form. It identifies the application with UN SDG 3 - Good Health & Well-being and provides a prominent "Get Recommendation" action. The top navigation provides direct access to Home, Advisor, Tips, BMI, Water and About sections.

4.8.2 IMPLEMENTATION - FOOD ADVISOR INPUT

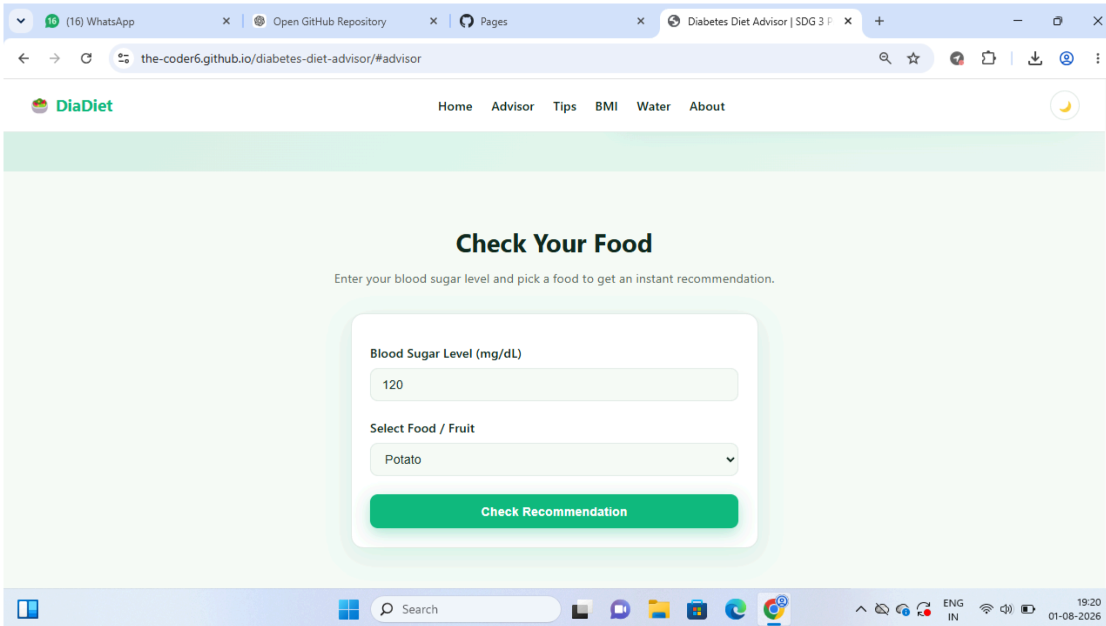

Figure 4.2: Food advisor input form

The food advisor asks for a blood sugar level in mg/dL and allows the user to select a food or fruit. The "Check Recommendation" button triggers the client-side logic. The form uses a minimal number of fields so that the interaction remains quick and understandable.

4.8.3 IMPLEMENTATION - RECOMMENDATION RESULT

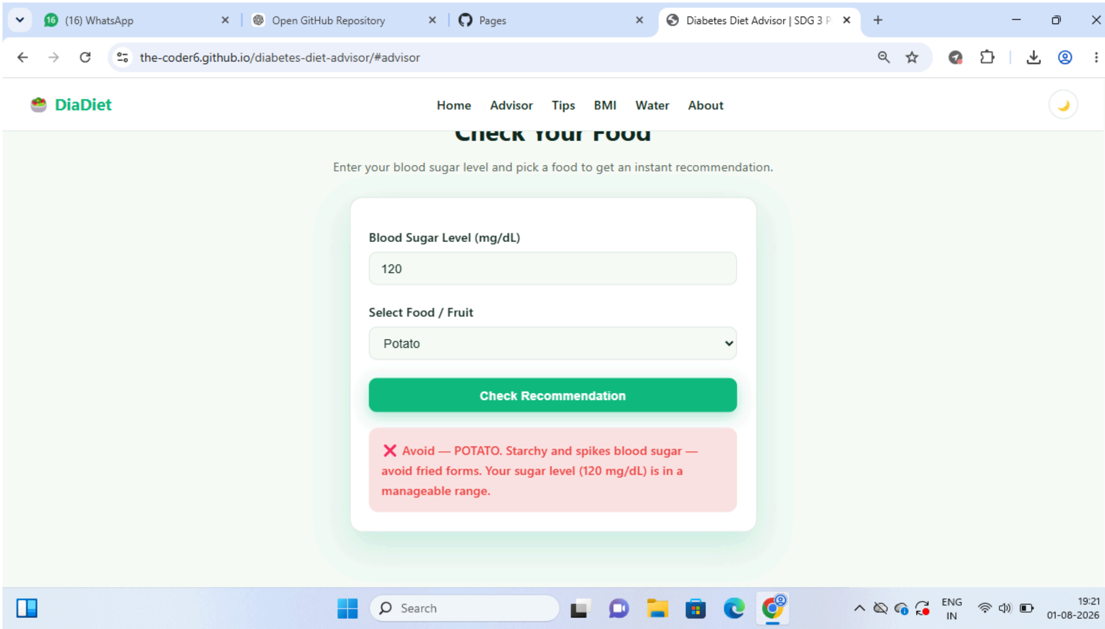

Figure 4.3: Example recommendation result for potato

The screenshot demonstrates a working result state. For the displayed example, a blood sugar value of 120 mg/dL and Potato are selected, after which the interface displays an "Avoid" recommendation and an explanatory message. This verifies that the page responds dynamically to user interaction.

The result should be treated as educational project output, not as medical advice. A production health system would require validated nutritional data, clinical review, individualized context and stronger safety controls.

4.8.4 IMPLEMENTATION - HEALTHY DIET TIPS

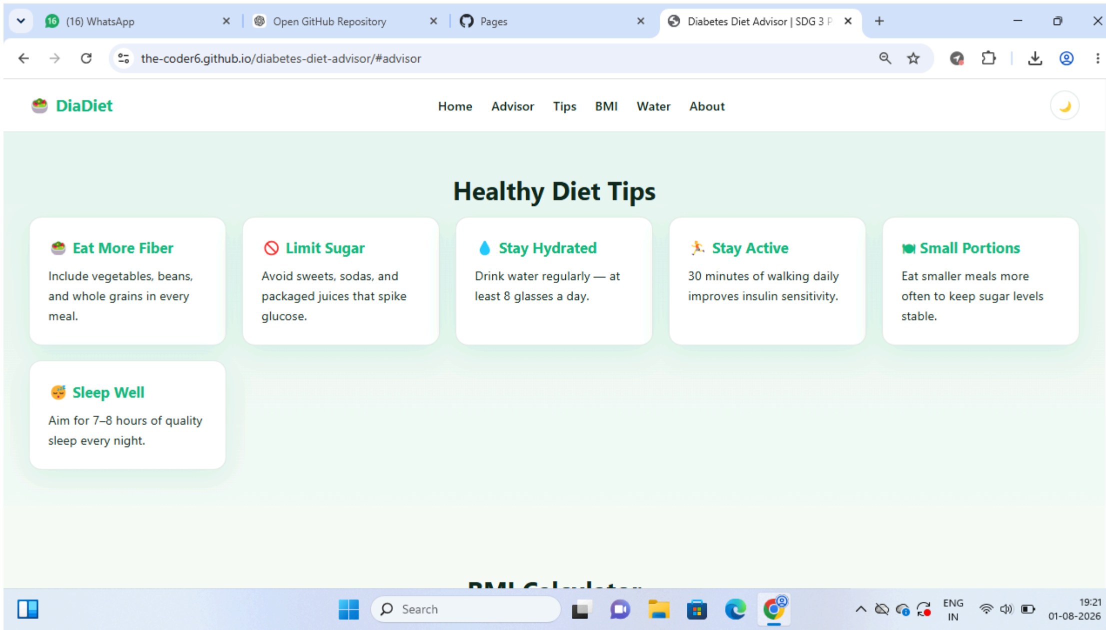

Figure 4.4: Healthy diet tips section

The Healthy Diet Tips section presents short, readable cards covering fiber intake, limiting sugar, hydration, activity, smaller portions and sleep. The card layout reduces visual complexity and makes the information easy to scan. This section complements the food advisor by encouraging broader wellness awareness.

4.8.5 IMPLEMENTATION - BMI CALCULATOR

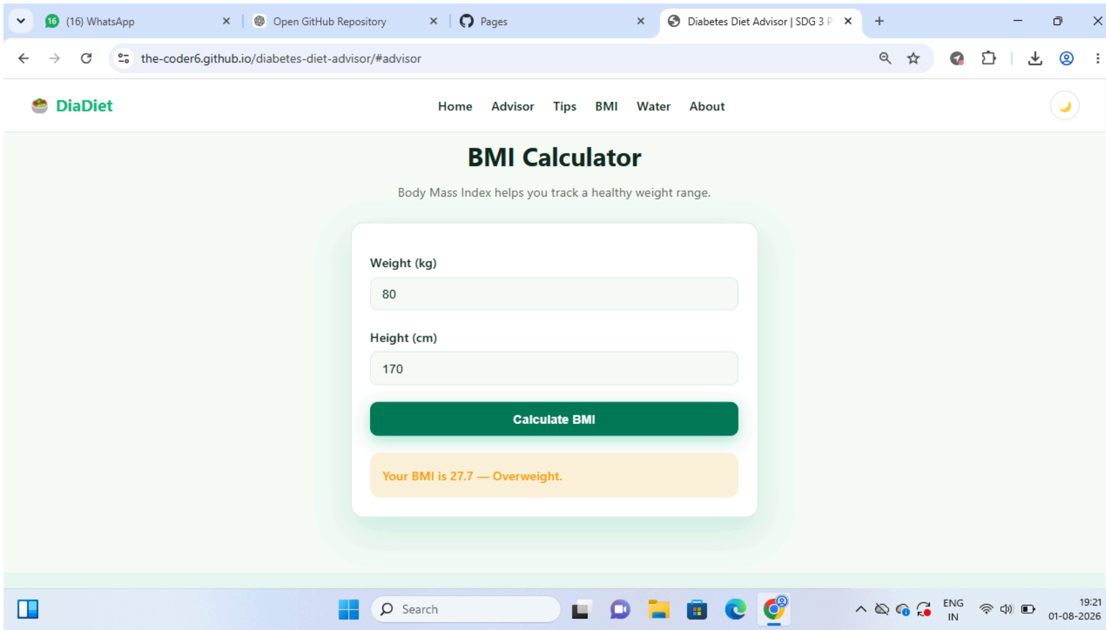

Figure 4.5: BMI calculator with example output

The BMI calculator accepts weight in kilograms and height in centimeters. The screenshot shows an example with 80 kg and 170 cm, producing a displayed BMI of 27.7 and the category "Overweight." The feature demonstrates numeric input handling, calculation and dynamic output presentation in JavaScript.

4.8.6 IMPLEMENTATION - WATER INTAKE TRACKER

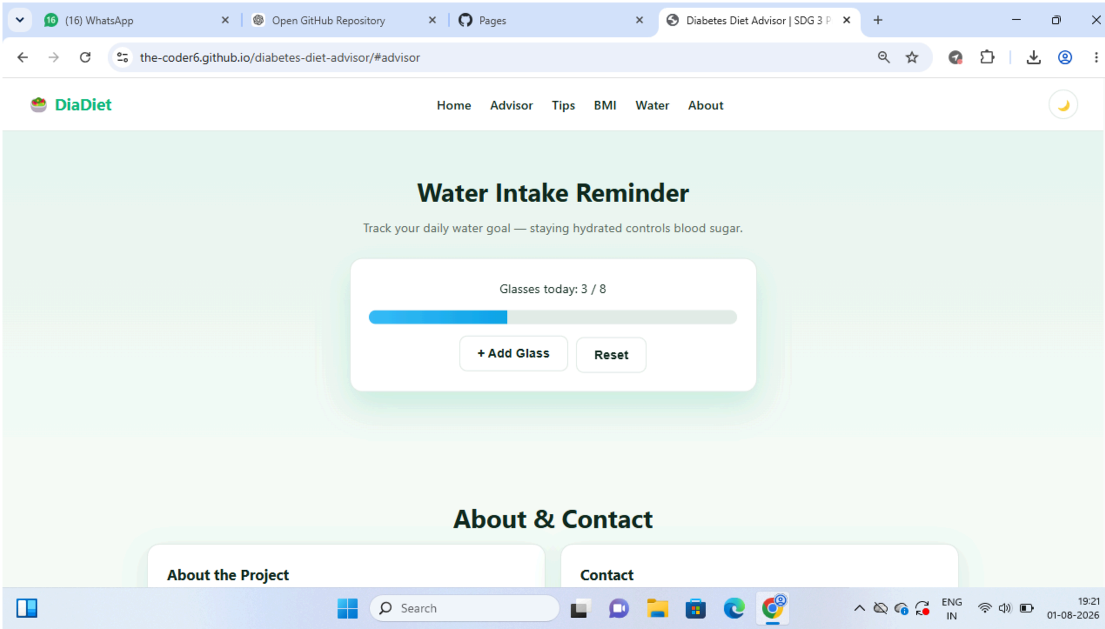

Figure 4.6: Water intake reminder and progress display

The water-intake module displays progress toward a daily goal of eight glasses. The user can add a glass or reset the counter. A progress bar provides immediate visual feedback. This feature demonstrates simple state tracking within the browser and extends the project beyond a single recommendation form.

4.8.7 IMPLEMENTATION - ABOUT & CONTACT

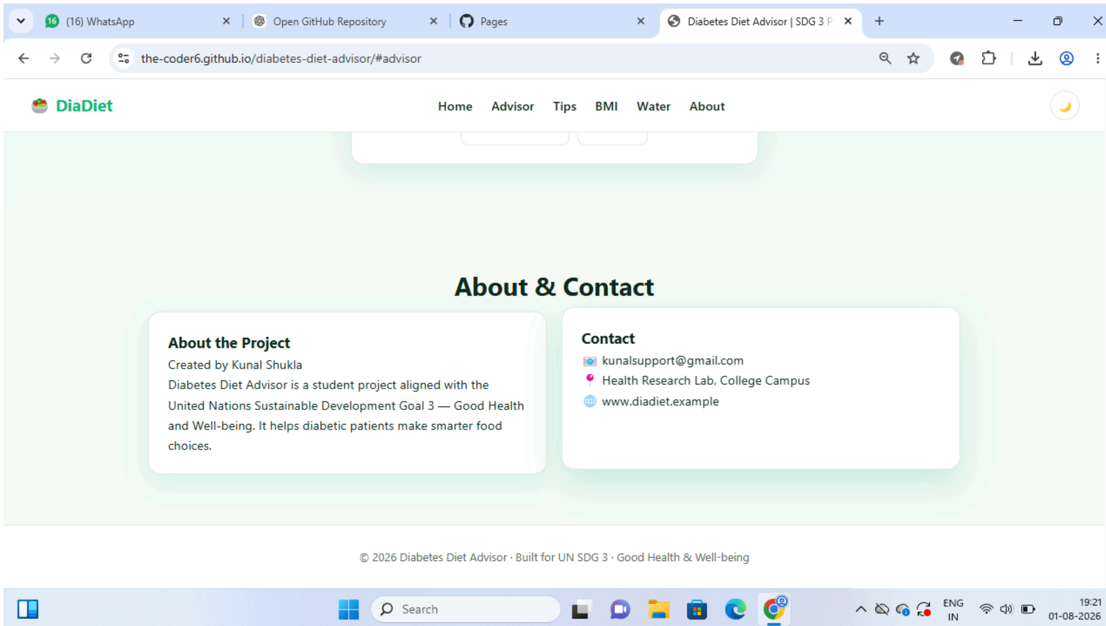

Figure 4.7: About and contact section

The About section identifies the project as created by Kunal Shukla and explicitly states its alignment with United Nations Sustainable Development Goal 3 - Good Health and Well-being. It describes the project as a student project intended to help users make smarter food choices. This section also improves transparency by communicating project ownership and purpose.

4.9 EXPECTED OUTCOMES AND LEARNING

Project Outcomes

- A functional and publicly accessible Diabetes Diet Advisor web application.
- A clear food-advisor interaction with immediate result display.
- Working BMI and water-intake utility modules.
- Responsive, organized health-oriented user interface.
- Public GitHub repository containing the project source.
- Public deployment using GitHub Pages.
- Documented alignment with UN SDG 3.

Internship Learning Outcomes

The internship materials show a progression from orientation through seven masterclasses, mentoring, assessment, career-readiness and hackathon activities. Masterclass 1 specifically covered Prompt Engineering for AI. The supplied learning material defined prompt engineering as the practice of designing and refining instructions for AI systems and introduced techniques such as zero-shot prompting, few-shot prompting and the Role-Goal-Context (RGC) framework.

For this project, the most important outcome was learning to move from a problem statement to a usable product: selecting a problem, defining features, implementing a front-end interface, testing interactions, publishing the application and documenting the result. The student also reports completing the Skill India assessment, although separate documentary proof was not available for inclusion.

Limitations and Future Enhancements

- Add a validated nutrition dataset and evidence-based food metadata.
- Introduce user profiles and persistent storage.
- Add clinician-reviewed safety disclaimers and escalation guidance.
- Improve accessibility and automated testing.
- If appropriate and validated, add an AI/ML component for personalization rather than labeling rule-based logic as AI.
- Add backend APIs and secure authentication for advanced versions.

4.10 CERTIFICATES AND COMMUNICATION PROOF

The final academic submission should include the internship completion certificate and any available communication proof. The student supplied the completion certificate and internship offer-letter information during preparation of this report. The student also stated that the Skill India assessment and internship activities were completed, but no separate Skill India certificate or additional proof was available.

IILM University's internship submission email requires the Internship Report PDF, Internship Presentation PDF and Internship Completion Certificate PDF to be uploaded to a public GitHub repository using the prescribed repository naming convention.

5. BIBLIOGRAPHY / REFERENCES

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